THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · NCERT-TYPE FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. In a survey of 500 families, 140 owned a two-wheeler. The probability that a family chosen at random owns a two-wheeler is: 1

(a) $\frac{7}{25}$ **(b)** $\frac{18}{25}$ **(c)** $\frac{8}{25}$ **(d)** $\frac{7}{26}$

2. Which term of the sequence $t_n = 3n + 2$ is 53? 1

(a) 17th term **(b)** 16th term **(c)** 18th term **(d)** 19th term

3. The proposition "If a number is divisible by 6, then it is divisible by 3" is true. Its converse "If it is divisible by 3, then a number is divisible by 6" is: 1

(a) False **(b)** True only for some values

(c) It cannot be decided **(d)** True

4. The average of 14 numbers is 31. When a number 61 is included, the average of all 15 numbers becomes: 1

(a) 33 **(b)** 32 **(c)** 35 **(d)** 34

5. The divisor algorithm applied to 44 produces a list with how many entries? 1

(a) 5 **(b)** 6 **(c)** 8 **(d)** 7

6. In a parallelogram, two opposite angles are $(2x + 11)^\circ$ and $(x + 18)^\circ$. The value of x is: 1

(a) 9 **(b)** 8 **(c)** 6 **(d)** 7

7. The y-intercept of the line $x + 6y = 24$ is: 1

(a) 5 **(b)** 4 **(c)** 6 **(d)** 3

8. Take $\pi = \frac{22}{7}$. The total surface area of a cylinder of radius 3.5 cm and height 27 cm is: 1

(a) 603 cm^2 **(b)** 739 cm^2 **(c)** 807 cm^2 **(d)** 671 cm^2

9. The midpoint of the segment joining $(-4, 6)$ and $(8, -2)$ is: 1

(a) $(-2, -2)$ **(b)** $(2, 2)$ **(c)** $(2, -2)$ **(d)** $(-2, 2)$

10. The value of $3x + 5$ at $x = -2$ is: 1

(a) -1 **(b)** -2 **(c)** 0 **(d)** 1

11. An irrational number between 2 and 3 is: 1

(a) $\sqrt{3}$ **(b)** $\sqrt{11}$ **(c)** $\sqrt{10}$ **(d)** $\sqrt{5}$

12. $(x + 7)(x - 3) =$ 1

(a) $x^2 + 10x - 21$ **(b)** $x^2 + 4x - 21$ **(c)** $x^2 + 4x + 21$ **(d)** $x^2 - 4x - 21$

13. The chord AB subtends 74° at the centre O. The angle subtended at a point on the major arc is: 1

(a) 27° **(b)** 47° **(c)** 37° **(d)** 57°

14. The area of a circle with circumference 88 cm (take $\pi = \frac{22}{7}$) is (cm^2): 1

(a) 617 **(b)** 615 **(c)** 618 **(d)** 616

15. A letter is chosen at random from the word EDUCATION. The probability that it is a vowel is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{4}{9}$ **(c)** $\frac{5}{9}$ **(d)** $\frac{2}{3}$

16. A sequence has $t_1 = 4$ and $t_{n+1} = t_n + 3$. Then t_6 equals: 1

(a) 20 **(b)** 19 **(c)** 21 **(d)** 18

17. Which of the following is a counterexample to the statement "If $a < b$, then $a^2 < b^2$"? 1

(a) $a = 3,\ b = 6$ **(b)** $a = 2,\ b = 5$ **(c)** $a = -3,\ b = 2$ **(d)** $a = 1,\ b = 4$

18. Marks of 70, 65 and 75 are given weights 6, 6 and 3 respectively. The weighted average is: 1

(a) 69 **(b)** 68 **(c)** 70 **(d)** 71

19. By Euclid's algorithm, gcd$(90, 82)$ is: 1

(a) 2 **(b)** 4 **(c)** 3 **(d)** 1

20. Two adjacent angles of a parallelogram are in the ratio $1 : 8$. The smaller angle is: 1

(a) 10° **(b)** 40° **(c)** 20° **(d)** 30°

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. A die was thrown 100 times with frequencies of $1, \ldots, 6$ being 15, 22, 17, 16, 14, 16. Find the experimental probability of getting an even number. 2

22. A sequence is defined by $t_n = 3n + 1$. Find t_8 and decide whether 37 is a term. 2

23. Write the converse of "If a number is divisible by 6, then it is divisible by 3". Is it true? 2

24. The average marks of 27 students in a test is 67 and that of 45 students is 35. Find the average marks of all 72 students together. 2

25. Apply Euclid's subtraction algorithm to find gcd(54, 30). Write the sequence of pairs and count the subtractions. 2

26. In a parallelogram ABCD, $\angle A = 88^\circ$. Find $\angle B$, $\angle C$ and $\angle D$. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. Write the converse of "If a number is divisible by 6, then it is divisible by 3". Is the converse true? Justify. 3

28. 110 mL of a 38% sugar solution is mixed with 220 mL of a 44% sugar solution. Find the percentage concentration of the mixture. 3

29. Apply Euclid's subtraction algorithm to find gcd(92, 72). Write the sequence of pairs and count the subtractions. 3

30. The opposite angles of a parallelogram are $(2x + 8)^\circ$ and $(x + 16)^\circ$. Find x and all four angles. 3

31. Find k if the points $(-3, 1)$, $(-1, -3)$ and $(1, k)$ lie on one straight line. 3

OR

Take $\pi = \frac{22}{7}$. A cone has radius 14 cm and height 48 cm. Find its slant height, curved surface area and volume. 3

32. Find the point on the x-axis which is equidistant from $A(-1, 4)$ and $B(5, -4)$. 3

OR

The line $y = 2x + k$ has slope 2 and passes through $(3, 11)$. Find k and write the line. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. Three sections have 22, 26 and 38 students with average marks 60, 47 and 44 respectively. Find the average marks of all the students. 5

34. Trace: x = 5; repeat 5 times: x = 2x − 2. Write the value of x after each repetition and the final value. 5

35. In $\triangle ABC$, D is the midpoint of AB and the line through D parallel to BC meets AC at E. $BC = 38$ cm and $AB = 28$ cm. Find DE, AD and show that E is the midpoint of AC. 5

36. Draw the graph of $2x + y = 3$ using a table of values for $x = -1, 0, 1, 2$. State its slope and where it cuts the y-axis. 5

OR

A rectangular sheet 44 cm long and 25 cm wide is rolled so that its length forms the circumference of the base of a cylinder of height 25 cm. Find the radius, the curved surface area and the volume (Take $\pi = \frac{22}{7}$). 5

37. The vertices of a triangle are $A(0, 0)$, $B(-9, 12)$, $C(-9, 0)$. Find the lengths of its three sides and its perimeter. 5

OR

For $y = 4x - 12$: 5

(i) find y at $x = 0, 1, 2$;

(ii) find the point where the graph crosses the x-axis;

(iii) state the slope and say whether the graph rises or falls from left to right.

38. Simplify $(5 + \sqrt{3})(5 - \sqrt{3})$ and $(5 + \sqrt{3})^2$ and say which result is rational. 5

OR

Expand $(x + 2)^3$. 5

END OF QUESTION PAPER — Total: 80 Marks

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